



DPUBooleanGate

Computes basic logical operations.

1. Inputs:

In[1...20] Inputs for the logical operations.

2. Static parameters:

Channels The inputs *In1* to *InChannels* are included in the computation of the logical operation.

Gate Type of operation. Possible values: AND, OR, XOR. Default: OR

3. Outputs:

Out Result of the logical Operation.

Out_Not Negated result.

4. Functional principle:

The DPU performs the following computation:

$$Out = In1 \otimes In2 \otimes In_{channels}$$

$$Out_Not = \neg Out$$

Herewithin, the operator $\otimes$ represents one of the Boolean operations AND, OR and XOR.

5. Example:

Computation of *In1* XOR *In2*

```
DPUBooleanGate xor
{
    Computer = {LOCALHOST};
    Index = 5;
    Channels = 2;
    Gate = XOR;
};
```